Submittal Review Response

Project Name:	***Hilo WWTP Rehabilitation and Replacement Project Phase 1***
Submittal No.:	***16133-003.0***
Date:	***9/25/2025***

Client:	County of Hawaiʻi	**Carollo Project No.:**	203975
Contractor:	Nan, Inc.		
Submittal Name:	Duct Bank		
Reviewed By:	Skyler Caoua		

SUBMITTAL REVIEW		
Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.		
Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. Conduit Spacers:
    a. Cantex spacers are acceptable per Spec 16133-2.02.A.
    b. QwikDuct spacers may be considered “or equal” but require confirmation that they meet Section 16133-2.04.A (non-metallic, interlocking, rebar holder, spacing, and concrete encasement requirements) and demonstrates interlocking features and conduit spacing compliance per Specification Section 3.01.C for 2” minimum separation & separate medium voltage ducts a minimum of 7.5” on center.
2. Detectable Underground Marking Tape:
    a. Panduit is highlighted in the Specification sections 1.02 B. 2. & 2.02 B. but product data was not submitted as specified in Specification section 1633-1.02.B. . Please Provide Product data sheet showing compliance with 2.04.C of having aluminum core, 6” wide, red with black lettering in submittal.

3. Pull Line:
    a. Greenlee pull line is acceptable per Spec 16133-2.02.C.
4. Duct Seal:
    a. O-Z/Gedney Type DUX is acceptable per Spec 16133-2.02.D.
    b. Ideal duct seal approved as may be considered “or equal” but requires confirmation of UL listing and compliance with Section 16133-2.04.D (non-hardening, moisture-barrier, UL-listed for installed conductors) Contractor to provide confirmation that ideal duct seal is UL listed for use with conductors and matches the non, hardening, moisture-barrier requirements.
5. Missing from Submittal:
    a. Conduit product data required per Section 16133-2.03.A and cross-reference to Section 16130.

☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:** 16133.003.0
Submittal Title: Duct Banks ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier		
Spec: 16133	**Paragraph:** 1.02.B	
Authored By: Mass Electric		**Date Submitted:** 9/10/2025

Submittal Certification

Check Either (A) or (B):

☑ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: [signature]

Printed Name and Title: Mark McCarthy QC

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/9/2025
SPECIFICATION SECTION # 16133
SPECIFICATION Conduit
PARAGRAPH 1.02.B
DRAWING
SUBCONTRACTOR Mass Electric
SUPPLIER
MANUFACTURER Various
CERTIFIED BY CQCM or Designee : [signature]

Submittal

SUB Reference No : **MECI-SUB-0008**

Status: OUTSTANDING

For Action:	Darrin Lee, MECI David Wieseler, MECI
Project:	HILO WWTP REHABILITATION AND REPLACEMENT PROJECT
Subject:	**16133 - DUCT BANK**

Submittal for underground conduit systems and duct bank accessories, including conduit spacers, detectable underground marking tape, pull line, and duct seal.

Submitted For: Approval

Specification Reference : 16133 - DUCT BANKS

Paragraph No.: 16133 - 1.02.B

Description : Submittal for underground conduit systems and duct bank accessories, including conduit spacers, detectable underground marking tape, pull line, and duct seal.

Discipline: ELEC **Area:** 00

Attachments:	HWWTP - 16133 - DUCT BANK.pdf	
SUB by:	Hannah Anderson, MECI	On: 10 September 2025
Created By:	Hannah Anderson, MECI	On: 10 September 2025, 08:05:21 AM -10:00
Last Edited By:	Hannah Anderson, MECI	On: 10 September 2025, 08:05:21 AM -10:00

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01 - QC/Project Engineer Review

Approved without comment.

QC/Project Engineer Review by:	David Wieseler, MECI	On: 10 September 2025
Created By:	David Wieseler, MECI	On: 10 September 2025, 08:07:45 AM -10:00
Last Edited By:	David Wieseler, MECI	On: 10 September 2025, 08:07:45 AM -10:00

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02 - Project Manager Review

Project Manager Review by:		On:
Created By:		On: ,
Last Edited By:		On: ,

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03 - Response

Response by:	On:
Created By:	On: ,
Last Edited By:	On: ,

04 - Project Manager/Engineer Closeout

Project Manager/Engineer Closeout by:	On:
Created By:	On: ,
Last Edited By:	On: ,

	MASS ELECTRIC CONSTRUCTION COMPANY PRODUCT DATA SUBMITTAL			
mass. electric construction company	**REVISION**	DATE	PREPARED BY	APPROVED BY
	0	**09/10/2025**	**Hannah Anderson**	**David Wieseler**

HILO WASTEWATER TREATMENT PLANT REHABILITATION AND REPLACEMENT PROJECT

HILO WWTP
MASS ELECTRIC HAWAII

SPECIFICATION - 16133

DUCT BANKS

DOCUMENT REVISION LOG						
Revision Number	**Revision Date**	**Description**	**Approvals**			
			PE INITIAL		**PM INITIAL**	
0	09/10/2025	Issued for Review	**DCW**	**09/10/2025**		

COMMENTS AND CLARIFICATIONS

- **Conduit Spacers (Spec 2.02.A)**
  As an alternate to the duct bank spacer specified, this submittal proposes the use of the QwikDuct spacer system. The alternate is submitted for consideration based on improved constructability and enhanced safety.
- **Duct Seal (Spec 2.02.D)**
  This submittal includes Ideal duct seal as an alternate to the specified product, due to availability.

SPECIFICATION – 16060
GROUNDING AND BONDING

TABLE OF CONTENTS

9/10/2025

SECTION 16133

DUCT BANKS

PART 1 GENERAL

1.01 SUMMARY

A. Section includes:
   1. Electrical underground duct banks.
   2. Duct bank installation requirements.

1.02 SUBMITTALS

A. Furnish submittals as specified in Section 01330 - Submittal Procedures.

B. Product data:
   1. PVC conduit spacers.
   2. Detectable underground marking tape.
   3. Pull line.

C. Shop drawings:
   1. Submit site plan drawings of duct banks including underground profiles indicating all underground utilities.
   2. Submit cross section of each duct bank with dimensions.
   3. For duct bank routings crossing under building footers or foundations alternative to designed routings indicated on the Drawings:
      a. Submit shop drawings detailing the new building footer crossing locations and plan drawings labeling all equipment to be installed on top of the new routing for approval by the project Structural Engineer.

1.03 PROJECT OR SITE CONDITIONS

A. As specified in Section 01850 - Design Criteria.

B. Field conditions and related requirements:
   1. Underground water table may be near or above the location of new duct banks.
   2. Include cost for necessary dewatering, and cleaning equipment to perform work in underground duct banks, pullboxes and manholes, before installation.

1.04 WARRANTY

A. As specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

A. Duct bank sections indicated on the Drawings are an initial arrangement and may need to be modified due to existing site conditions, existing underground utilities, and infrastructure.
   1. Reorganize duct bank section with approval of Engineer.
   2. Make changes required to accommodate duct bank configuration and routing changes due to field conditions.
      a. Where changes in a duct bank configuration extend to a manhole or handhole, coordinate manhole/handhole blockouts and size with the new configuration.

B. Provide location and protection of existing underground utilities, duct bank, trenching, forming, rebar, spacers, conduit, concrete, backfill, and compaction necessary for the complete installation of the duct banks.

C. Provide reinforced concrete duct banks for all conduits installed below grade, on the site, below structures, or in contact with the earth, unless otherwise indicated on the Drawings.

2.02 MANUFACTURERS

A. Conduit spacers:
   1. One of the following or equal:
      a. Cantex.
      b. Osburn Associates, Inc.

B. Detectable underground marking tape:
   1. One of the following or equal:
      a. Blackburn Manufacturing Co.
      b. Panduit.

C. Pull line:
   1. One of the following or equal:
      a. Arnco.
      b. Greenlee.
      c. Osburn Associates, Inc.

D. Duct seal:
   1. The following or equal:
      a. O-Z/Gedney type DUX.

2.03 MATERIALS

A. Provide conduit as specified in Section 16130 - Conduits.

B. Provide reinforcing steel as specified in Section 03200 - Concrete Reinforcing:
   1. Provide minimum Number 4 reinforcing steel.

C. Concrete:
   1. Mix requirements as specified in Section 03300 - Cast-in-Place Concrete.
   2. Provide a red-oxide conduit encasement coloring agent as specified in Section 03300 - Cast-in-Place Concrete.

2.04 MANUFACTURED UNITS

A. Conduit spacers:
   1. Provide conduit spacers recommended by the conduit manufacturer or specified above.
   2. Saddle type.
   3. Non-metallic, non-corrosive, non-conductive.
   4. Interlocking type:
      a. Vertical interlocking.
      b. Horizontal interlocking.
   5. Suitable for concrete encasement.
   6. Molded-in rebar holder.
   7. Accommodates 2-inch through 6-inch conduit sizes.
   8. Relieves the conduit from both horizontal and vertical stresses.

B. Pull line:
   1. Minimum 1/4-inch wide, flat design.
   2. Polyester.
   3. Minimum pulling strength 1,200 pounds.

C. Detectable marking tape:
   1. Provide a detectable tape, locatable by a cable or metal detector from above the undisturbed grade.
   2. Aluminum core laminated between polyethylene film.
   3. 6-inch-wide red tape imprinted with black lettering stating "CAUTION - BURIED ELECTRIC LINE BELOW" or equivalent.

D. Duct seal:
   1. Non-hardening sealing compound.
   2. Flexible, can be applied by hand.
   3. UL Listed for use with installed conductors.

PART 3 EXECUTION

3.01 INSTALLATION

A. Duct banks:
   1. Install duct banks encased in concrete at least 24 inches below finish grade, unless otherwise indicated on the Drawings.
   2. Damage minimization:
      a. Conduit should not be left exposed in an open trench longer than is necessary.
      b. Protect all underground duct banks against damage during pouring of concrete or backfilling.

3. All plastic conduit fittings to be joined should be exposed to the same temperature conditions for a reasonable length of time before assembly.
4. Provide No. 4/0 American Wire Gauge bare copper ground wire the entire length of duct bank and bond to the grounding system at each end of the duct bank.
5. Install underground ducts to be self-draining:
    a. Slope duct banks away from buildings to manholes, handholes, or pullboxes.
    b. Slope duct banks uniformly from manholes, handholes, or pullboxes to manholes, handholes, or pullboxes or both ways from high points between manholes, handholes, or pullboxes.
    c. Slope a minimum of 1/4 inch per 10 feet.
6. Where new duct banks join to existing manholes, handholes, or pullboxes, make the proper fittings and fabricate the concrete envelopes to ensure smooth durable transitions, as indicated on the Drawings.
7. Install pull line in spare conduits:
    a. Provide adequate pull line at both ends of conduits to facilitate conductor pulling.
    b. Cap above ground spare conduit risers at each end with screw-on conduit caps.

B. Trenching:
1. Perform trenching as specified in Section 02318 - Trenching.
2. Trench must be uniformly graded with the bottom, rock free and covered with select material.
3. Whenever possible, use the walls of the trench as forms for concrete encasement:
    a. Forms are required where the soil is not self-supporting.
4. Damage occurring to existing ducts, conduits, cables, and other utilities during duct bank installation shall be remediated to the satisfaction of the Owner.

C. Duct spacing:
1. Separate conduits with manufactured plastic spacers using a minimum space between the outside surfaces of adjacent conduits of 2 inches, unless otherwise indicated on the Drawings:
    a. Separate medium voltage ducts a minimum of 7.5 inches on center.
2. Install spacers to maintain uniform spacing of duct assembly a minimum of 4 inches above the bottom of the trench during concrete pour. Install spacers on 8-foot maximum intervals:
    a. Due to some distortion of conduit from heat, and other means, it may be necessary to install extra spacers within the duct bank:
        1) Install the intermediate set of spacers within normal required spacing to maintain the proper horizontal clearance:
            a) Clearance is required to allow the proper amount of concrete to infiltrate vertically among the duct to ensure proper protection.
3. Spacers shall not be located at the center of a bend:
    a. Locate spacer in the tangent, free of the coupling on fabricated bends.
    b. Locate spacers midway between the tangent and the center bend on trench formed sweeps.

D. Terminating:
   1. Use bell ends in duct at entrances into cable vaults.
   2. Make conduit entrances into cable vaults tangential to walls of cable vault.
   3. Form trapezoidal transitions between duct bank and cable vaults as needed in order to ensure adequate cable bending radius for the duct bank-to-vault transition.
   4. Install duct seal in all conduits including spare conduits, at both ends, entrance to manholes/handholes , and building/equipment stub-ups. Form by hand to conduit and around cables to develop moisture barrier.
   5. New manhole or handhole applications, provide a single opening or "window" per duct bank, sized to accommodate the duct bank envelope.

E. Concrete:
   1. Install concrete as specified in Section 03300 - Cast-in-Place Concrete.
   2. Provide tie wires in accordance with Section 03200 - Concrete Reinforcing to prevent displacement of the conduits during pouring of concrete:
      a. Tie wire shall not act as a substitute for spacers.
   3. Install minimum 3-inch cover around conduit and rebar.
   4. Consolidation of encasement concrete around duct banks shall be by hand puddling. Mechanical vibrators are acceptable for use outside of the rebar cage.
   5. Conduit is subject to temperature rise. As concrete cures, allow the free end to expand by pouring the concrete from the center of the run or from one tie in point.

F. Marking tape:
   1. Install a detectable marking tape 12 inches above the duct bank the entire length of the duct bank.

G. For conduit installations beneath building slabs:
   1. Duct banks shall be continued under building slabs to the final destination of the conduits.
      a. Construct separate duct banks as required.
      b. Concrete for encasement under building slabs need not be colored red.
      c. For duct banks crossing under building footers or foundations, install the top of the duct bank a minimum of 6 inches below the footer.
      d. Where duct banks enter through building walls, foundation walls, stem walls, etc. make connections as indicated on the Drawings.
      e. Where duct banks terminate with conduit risers entering building walls, install an expansion/deflection fitting or a flat-wise elbow (elbow parallel to building wall) in order to accommodate differential movement between the conduits and structure.

H. Restore all surfaces to their original condition as specified in Section 02952 - Pavement Restoration and Rehabilitation, unless otherwise specified.

3.02 COMMISSIONING

A. As specified in Section 01756 - Commissioning.

3.03 CLEANING

A. Clean conduits of dirt and debris by use of an appropriately sized steel mandrel no less than 1/2 inch smaller than the inside diameter of the conduit.

3.04 PROTECTION

A. Provide shoring and pumping to protect the excavation and safety of workers.

B. Protect excavations with barricades as required by applicable safety regulations.

END OF SECTION

1.0 CONDUIT SPACERS

SPECIFICATION – 16133-2.02.A
DUCT BANKS

CANTEX®

NEW & IMPROVED BASE SPACERS

- IMPROVED STRENGTH & RIGIDITY
- IMPROVED CRUSH TEST RATING
- IMPROVED STABILITY

WE LISTENED!

Evolving customer needs, technological improvements, and customer input are a few factors that led CANTEX to the decision to improve our base spacers. At CANTEX, we value what our customers tell us, so we continually make changes when the end-users think we can improve our PVC electrical products. After hearing from customers who thought certain design features of our PVC spacers could be improved, CANTEX began working to improve the design of our base spacers for strength and durability while maintaining all the existing functionality and cost advantages.

Product Improvements to Base Spacer:

- Strengthened webbing and cross beam to improve strength and rigidness
- Significantly improved the crush test rating
- Added ¼ inch to each side of the base of all trade sizes for better stability

2.04.A.5

Product Description:

Base Spacers are used to create concrete encased banks that are strong and stable with consistent duct separation.

- Vertical and horizontal interlocking for fast secure field assembly 2.04.A.4
- Wide base plate provides solid support for the heaviest loads
- For use with all EB & DB and schedule 40 & 80 conduit when encased in cement
- Only base and intermediate components are needed - no banding necessary
- Material is Polystyrene 2.04.A.3
- Made in USA

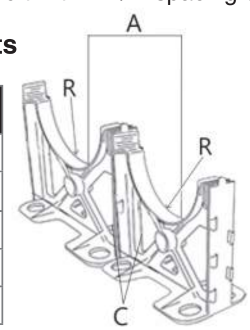
Spacer Separation Requirements:

Base spacers provide a 3" separation between bottom row of ducts & trench floor, except for 4" x 1" which is 1-3/4". To determine the trade size needed for your spacers, look at the size of the pipe and the size of the separation needed. For example, the 2 x 1-1/2 in. Base Spacer can handle up to 2" diameter conduit with 1-1/2" spacing between conduit runs.

NEC 70 Spacer Requirements

Conduit Trade Size	Maximum Spacing Between Supports
1/2 to 1	3 ft.
1-1/4 to 2	5 ft.
2-1/2 to 3	6 ft.
3-1/2 to 5	7 ft.
6	8 ft.

2.04.A.7



Nominal Size	1" Separation			1-1/2" Separation			2" Separation			3 Separation		
	R	A	C	R	A	C	R	A	C	R	A	C
2				1.2	4.0	1.5	1.2	4.5	2.0	1.2	5.5	3.0
3				1.8	5.1	1.5	1.8	5.6	2.0	1.8	6.6	3.0
4	2.3	5.6	1.0	2.3	6.1	1.5	2.3	6.6	2.0	2.3	7.6	3.0
5				2.9	7.3	1.5	2.9	7.9	2.0	2.9	8.9	3.0
6				3.3	8.2	1.5	3.4	8.7	2.0	3.4	9.8	3.0

CANTEX INC.
301 Commerce Street, Suite 2700
Fort Worth, Texas 76102-4127

www.cantexinc.com
info@cantexinc.com

817.215.7000

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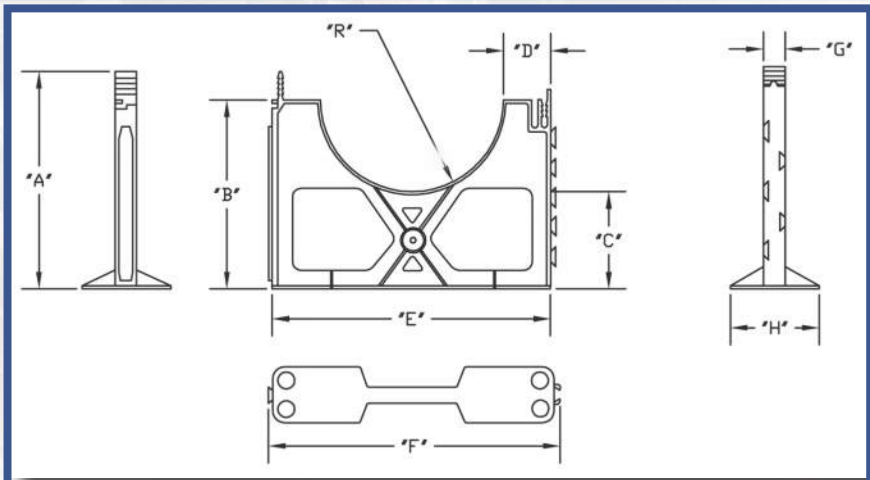

Base Spacer Sizes

Part Number	Size	Carton Qty.	"A"	"B"	"C"	"D"	"E"	"F"	"G"	"H"	"R"
5335961	2 x 1-1/2	100	4.875	4.250	3.000	.750	4.000	4.250	1.000	3.000	1.250
5335977	2 x 2	100	4.875	4.250	3.000	1.000	4.500	4.750	1.000	3.000	1.250
5335979	2 x 3	100	4.875	4.250	3.000	1.500	5.500	5.750	1.000	3.000	1.250
5335963	3 x 1-1/2	90	5.500	4.750	3.000	.750	5.125	5.375	1.000	3.000	1.813
5335964	3 x 2	80	5.500	4.750	3.000	1.000	5.625	5.875	1.000	3.000	1.813
5335975	3 x 3	70	5.500	4.750	3.000	1.500	6.625	6.875	1.000	3.000	1.813
5335971	4 x 1	80	5.938	5.250	3.000	.500	5.625	5.875	1.000	3.000	2.313
5335968	4 x 1-1/2	75	5.938	5.313	3.000	.750	6.125	6.375	1.000	3.000	2.313
5335967	4 x 2	70	5.938	5.313	3.000	1.000	6.625	6.875	1.000	3.000	2.313
5335969	4 x 3	65	5.938	5.250	3.000	1.500	7.625	7.875	1.000	3.000	2.313
5335980	5 x 1-1/2	60	6.500	5.813	3.000	.750	7.813	7.438	1.000	3.000	2.875
5335976	5 x 2	60	6.500	5.813	3.000	1.000	7.688	7.938	1.000	3.000	2.875
5335978	5 x 3	50	6.500	5.813	3.000	1.500	8.688	8.938	1.000	3.000	2.875
5336039	6 x 1-1/2	50	7.000	6.375	3.000	.750	8.250	8.500	1.000	3.000	3.375
5336051	6 x 2	50	7.000	6.375	3.000	1.000	8.750	9.000	1.000	3.000	3.375
5336040	6 x 3	50	7.000	6.375	3.000	1.500	9.750	10.000	1.000	3.000	3.375

Dimensions are nominal

CANTEX INC.
301 Commerce Street, Suite 2700
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www.cantexinc.com
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OR DEATHS

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TENSILE STRENGTH
AT BREAK

OVER 12X FASTER
THAN SPACERS
& CHAIRS

NECA SHOW STOPPER 2000

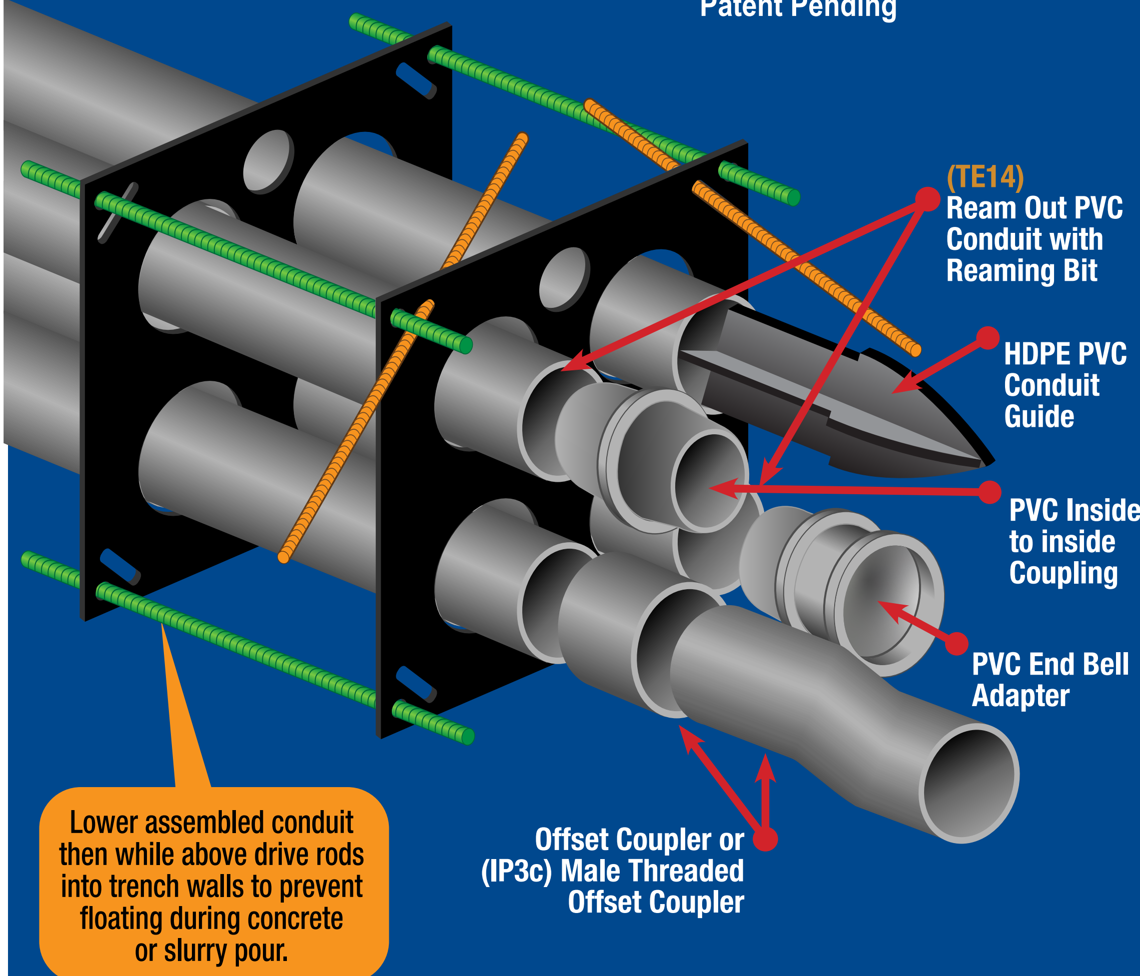
c UL US

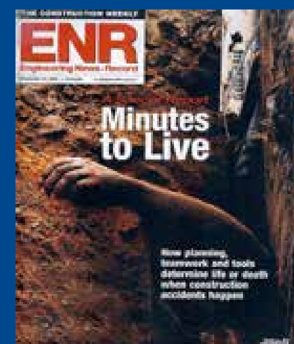
QwikDuct

HDPE CUSTOM TEMPLATES

Patent Pending

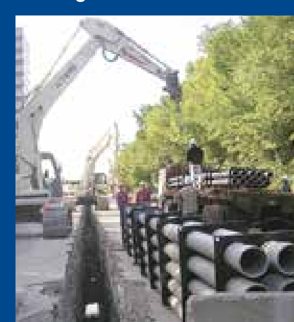
IP1





Eliminates dangerous & expensive time working in trench strengthening weak spacers & chairs with tie wire or other methods.

Lowering assembled conduit from above allows other savings – Digging a narrower trench for less excavation, concrete or slurry, shoring and backfill. The savings are huge!



ALSO SUITABLE FOR TRANSFORMER & SWITCHGEAR STUB UPS

Submittals provided with all duct bank quotes

SP PRODUCTS, INC
Productivity and value thru ideas

MADE IN USA · UL · c UL US · PATENT PENDING · PATENTED

LABELS APPLY TO PRODUCTS ONLY WHERE SHOWN ABOVE

730 Pratt Boulevard • Elk Grove Village, IL 60007
P: (847) 593-8595 • F: (847) 593-8629 • Toll-free (800) 233-8595
E: info@spproducts.com • www.spproducts.com

IP1

Eighteenth Printing 7-1-2025

QwikDuct HDPE CUSTOM TEMPLATES

MADE IN USA

c UL US

WHEN ORDERING CUSTOMER MUST SPECIFY:

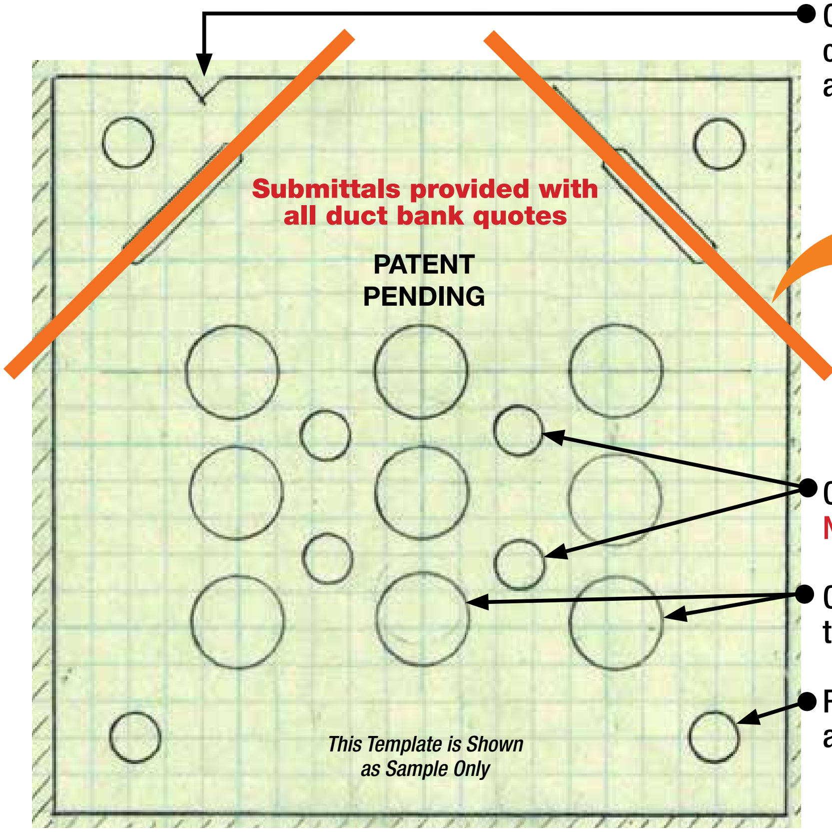
1. Material Thickness 3/8", 1/2", 3/4" and Load Rating Required (1/2" & ¾" are special order materials)
2. Size & Location of Rebar Holes
3. Spacing Between Conduit Openings
4. Rebar Stake Down Slots Required?
5. Diameter of Conduits
6. Footage Between Templates
7. Size & Location of Concrete Flow Holes

WHEN ASSEMBLING ABOVE THE TRENCH:

1. Install Conduits into Templates & Rebar if Required
2. Lower Assemblies into Trench
3. Insert Stake Down Rods into Angled Slots
4. Drive Stake Down Rods into Trench Walls from Above
5. Backfill with Concrete, Slurry, Sand or Gravel etc.

SEE WEBSITE VIDEO **IP1-2**

QwikDuct Templates can be used with all standardized trade size conduit specified per NFPA 70 NEC



Orientation notch for fast, correct alignment during assembly above trench.

Workers insert rods thru angled slots and drive rods into trench walls from above to prevent floating during concrete pour.

2.04.A.6

Concrete flow thru holes.
Note: specify size and location.

Conduit Holes Larger than Conduit OD (see **IP3**)

Rebar holes available at locations specified

2.04.A.3

QWIKDUCT HDPE CUSTOM TEMPLATES meet or exceed US Army Corps of Engineers requirement.

Customer must specify larger diameter holes to enable feeding bell ends of conduit from either direction.

PROVIDE SAFER & FASTER INSTALLATION PLUS ALLOW CLOSING TRENCH QUICKER TO REDUCE CHANCE OF DANGEROUS & COSTLY SOIL OR WATER PROBLEMS.

ALSO SUITABLE FOR TRANSFORMER & SWITCHGEAR STUB UPS

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Eighteenth Printing 7-1-2025

3.0 PULL LINE

SPECIFICATION – 16133-2.02.C
DUCT BANKS

Pulling Eye
8″ lock stitched on both ends of rope

40,000 Lbs. of Average Breaking Strength
Delivers superior strength

Durability
Ultra-low stretch, UV resistance and minimal water absorption

End-of-Pull Indicator
50′ orange-colored sections at each end are built for reliable pulling to promote user confidence

Green Tracers
Offer an additional layer of awareness in observing rope speed

60% Lighter*
Allowing for easier one-man transport and improved user ergonomics

CAT. NO.	LENGTH	DIAMETER	MAX. RATED CAPACITY		AVG. BREAKING STRENGTH		COMPATIBLE PULLER(S)
HPR300	300′	19/32″	10,000 lbs	44.5 kn	44,500 lbf	197.9 kn	6001, G6, UT-10, GX10
HPR600	600′						
HPR1200	1,200′						

*Compared to Greenlee's 7/8″ rope
**When used with 10K Puller

	CAT. NO.	DESCRIPTION
N	HPR300	300′ High Performance Rope
N	HPR600	600′ High Performance Rope
N	HPR1200	1,200′ High Performance Rope

N = NEW

Description

This document is to provide information about the selection, inspection, and maintenance of Greenlee pulling rope used with Greenlee pullers. The contents of this document is to be used as a supplement to your employer's and the industry's best practices and procedures regarding the care and inspection of pulling rope.

Greenlee pulling rope is white with a green tracer with factory spliced eyes at both ends. Read the puller manual and consult the charts in the Greenlee product catalog to help choose the correct rope and accessories for the puller.

GREENLEE PULLING ROPE FEATURES	
DOUBLE BRAIDED PULLING ROPE	**HIGH PERFORMANCE PULLING ROPE**
Construction: Braided Core Encased in Braided Cover **Material:** Polyethylene/ Nylon Composite **Storage Temp:** Up to 350 °F (176.7 °C) **Melting Point:** 460 °F (237.8 °C)	**Construction:** 12 Strand Single Braided Rope **Material:** High-Modulus Polyethylene **Storage Temp:** -125°F to 160 °F (-87°C to 71°C) **Melting Point:** 275 °F (135 °C)
	
High Strength Minimal Elongation Under Tension Protective Outer Sleeve of Rope Core Low Coefficient of Friction for Smoother Pulling and to Reduce Cable Damage. 8" Lock Stitched Pulling Eyes on Both Ends Flexible And Easy to Handle	Ultra-High Strength Minimal Elongation Under Tension UV Resistant Coating Lower Coefficient of Friction than Double Braided Rope for Faster Pulling. 8" Lock Stitched Pulling Eyes on Both Ends Flexible And Easy to Handle 50 Ft. Orange Dyed Rope at Each End Compressible On Capstans for Maximum Wraps

<table>
<tr><th colspan="11">GREENLEE PULLING ROPE SPECIFICATIONS*</th></tr>
<tr><th rowspan="2">ROPE TYPE</th><th rowspan="2">CAT. NO.</th><th rowspan="2">UPC NO.</th><th rowspan="2">LENGTH (ft)</th><th rowspan="2">DIAMETER (in)</th><th colspan="2">MAX. RATED CAPACITY</th><th colspan="2">AVG. BREAKING STRENGTH</th><th rowspan="2">COMPATIBLE PULLER(S)</th></tr>
<tr><th>lbf</th><th>kN</th><th>lbf</th><th>kN</th></tr>
<tr><td rowspan="14">DOUBLE BRAIDED PULLING ROPE</td><td>450</td><td>783310237730</td><td>300</td><td rowspan="3">3/8</td><td rowspan="3">1,375</td><td rowspan="3">6.1</td><td rowspan="3">5,500</td><td rowspan="3">24.5</td><td rowspan="3">805</td></tr>
<tr><td>451</td><td>783310237754</td><td>600</td></tr>
<tr><td>452</td><td>783310237761</td><td>1,200</td></tr>
<tr><td>455</td><td>783310237747</td><td>300</td><td rowspan="2">1/2</td><td rowspan="2">2,450</td><td rowspan="2">10.9</td><td rowspan="2">9,800</td><td rowspan="2">43.6</td><td rowspan="2">G3, UT2</td></tr>
<tr><td>456</td><td>783310237778</td><td>600</td></tr>
<tr><td>35285G</td><td>783310352839</td><td>300</td><td rowspan="3">9/16</td><td rowspan="3">4,000</td><td rowspan="3">17.8</td><td rowspan="3">16,000</td><td rowspan="3">71.2</td><td rowspan="3">UT6
640</td></tr>
<tr><td>35284</td><td>783310352846</td><td>600</td></tr>
<tr><td>35285G</td><td>783310352853</td><td>1,200</td></tr>
<tr><td>35098</td><td>783310350989</td><td>300</td><td rowspan="3">3/4</td><td rowspan="3">6,500</td><td rowspan="3">28.9</td><td rowspan="3">26,000</td><td rowspan="3">115.7</td><td rowspan="3">G6
6001</td></tr>
<tr><td>35100</td><td>783310351009</td><td>600</td></tr>
<tr><td>35101</td><td>783310351016</td><td>1,200</td></tr>
<tr><td>34136G</td><td>783310341369</td><td>300</td><td rowspan="3">7/8</td><td rowspan="3">8,000</td><td rowspan="3">35.6</td><td rowspan="3">32,000</td><td rowspan="3">142.3</td><td rowspan="3">UT-8</td></tr>
<tr><td>34137</td><td>783310341376</td><td>600</td></tr>
<tr><td>34138</td><td>783310341383</td><td>1,200</td></tr>
<tr><td rowspan="2">HIGH PERFORMANCE PULLING ROPE</td><td>HPR600</td><td>78331006671</td><td>600</td><td rowspan="2">19/32$^{\Delta}$</td><td rowspan="2">10,000</td><td rowspan="2">44.5</td><td rowspan="2">44,500</td><td rowspan="2">197.9</td><td rowspan="2">UT10
GX10</td></tr>
<tr><td>HPR1200</td><td>783310007340</td><td>1,200</td></tr>
<tr><td colspan="10">* All information for new, unused, and properly stored Greenlee Pulling Ropes
Δ Can be used in place of 7/8 in. or 3/4 in. ropes on appropriate Greenlee Pullers</td></tr>
</table>

4.0 DUCT SEAL

SPECIFICATION – 16133-2.02.D
DUCT BANKS

Compounds

Type DUX
Duct Sealing Compound

Catalog Number	Pound Package
DUX-1	1
DUX-5	5

Type DUX

Use:
An ideal waterstop and moisture barrier for sealing between electrical cables and conduit ducts. A standard for weather sealing service entries.

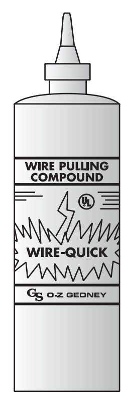
Features:
- A soft, fibrous, slightly tacky, non-hardening sealing compound easily applied by hand at all working temperatures. 2.04.D.1 2.04.D.2
- Gray color, clean and non-staining
- Can be painted
- Excellent cohesive and adhesive properties

Third Party Certification:
UL Listed: R-15320 2.04.D.3

Ordering Note:
When ordering DUX, specify number of units only, not lbs. For example: To order 50 lbs., specify 10 DUX-5 units.

Type WQ
Wire•Quick Wire Pulling Compound

Catalog Number	Unit
WQ-00	1 Qt. Can
WQ-01	1 Gal. Pail
WQ-05	5 Gal. Pail
WQ-55	55 Gal. Drum



Type WQ

Use:
"Wire Quick" lubricates wire and cable to substantially reduce the pulling tension required.

Features:
- It has excellent adhesion and will not crumble.
- Completely safe for use on all insulations… rubber, lead and plastics. and "WIRE QUICK" can be used with all types of conduit. It's non-corrosive and non-combustible.
- "WIRE QUICK", after a period of time in a conduit, will dry and leave a film. This acts as a lubricant in the event the wire or cable must be removed.
- Skin and clothing will not be harmed by Wire-Quick compound. It also washes easily from hands.

Third Party Certification:
UL Listed: E-76536

Ordering Note:
When ordering WIRE QUICK specify by number of units only, not gallons. For example: To order 50 gallons, specify 10 WQ-05 units.

Asbestos-Free Duct Seal
Technical Data Sheet

Asbestos Free Duct Seal is a gray, permanently soft, non-toxic compound which will adhere to most clean dry surfaces. Asbestos free Duct Seal will not adversely affect other plastic materials or corrode metals. It also has no adverse effect to human skin.

- FDA Approved: As listed in CFR, Title 21, being composed of ingredients acceptable for packaging and transporting food.
- U.S.D.A. Acceptable: Chemically acceptable to the U.S.D.A. for use in meat and poultry processing areas under Federal Inspection.
- Dielectric Strength: Approx. 100 volts per mil (ASTM D149-64). Resistivity: 1.5.1 @ 100°C...4.8 x 10^9..ohmcm 1.5.2 after 96 hrs. @ 115°..4.8 x 10^{10} (ASME power test code of 1965, Section 4.05).
- Chemical Resistance: Excellent resistance to water, alcohols, mild acids and bases.
- Vehicle Bleedout: None
- Non-Corrosive: Will not corrode Metals.
- Non-Irritant: No irritation to eyes or skin as listed in CFR, Title 16, "Appraisal of the safety of chemicals in food, drugs, and cosmetics.
- Paintability: Yes.

2.04.D.2

Uses:
Duct Seal is used primarily by the building and construction industry; and specifically by the electrical trade to seal around electrical boxes, flashings, and service mast entries, etc. It can be shaped by hand to any form and reused if necessary. Duct Seal has countless other applications in the refrigeration, heating and cooling, plumbing, and metal fabrication fields as well as being an excellent general purpose sealant around the home. It may be painted immediately after application and will not bleed through the dry paint.

Forms:
Duct Seal is individually wrapped and placed in plastic bags with convenient reusable clips to seal out dirt and moisture that might contaminate the product. It is available in one and five pound blocks, in cartons of 10.

2.04.D.1

Properties	Test Method	Value
Base		Non-drying synthetic polymers and oils
Fillers		Saturated mineral fillers and other inert ingredients
Specific Gravity	ASTM D71-72	1.65 to 1.7
Cone penetration	ASTM D217-52T	Load of 300 gm..5 sec. @ 25°F; 100 - 115 mm/10
Temperature usage range	25°F to +120° F	Recommended
Temperature tolerance range	-30°F to +175° F	Will not slump at +275°F

Description	Quantity	Part No.
1 lb. Block	Carton of 10	31-601
5 lb. Block	Carton of 10	31-605

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2.04.D.3

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